

FROM BRAIN STATES TO AMYLOID BURDEN: HIDDEN MARKOV MODELS OF RS-FMRI PREDICT SUBJECT-LEVEL PET SUVR

Rajesh R Nandy¹, Chendi Han Ph.D.², Dietmar Cordes²

1-University of North Texas Health, Fort Worth, TX. 2-Cleveland Clinic Lou Ruvo Center for Brain Health, Las Vegas, NV.

Introduction

Amyloid PET standardized uptake value ratio (SUVR) is a central biomarker of Alzheimer’s disease. However, PET imaging is expensive, not universally available, and exposes patients to ionizing radiation, limiting its use for large-scale screening or repeated monitoring. Scalable, non-invasive methods that can estimate continuous amyloid burden, rather than just amyloid positivity, would therefore be highly valuable for research and clinical care.

Resting-state functional MRI (rs-fMRI) is widely available and sensitive to early network-level alterations along the Alzheimer’s disease continuum. Prior work has shown that static functional connectivity is associated with amyloid deposition and can be used to predict amyloid status or related cognitive outcomes. Yet static connectivity collapses temporal information across the scan and may miss disease-relevant features encoded in the dynamics of large-scale brain networks.

Hidden Markov models (HMMs) provide a principled framework to characterize these fast dynamics, representing rs-fMRI as transitions among a finite set of recurrent brain states [1]. Subject-level HMM metrics such as fractional occupancy (FO), mean dwell time, and transition probabilities compactly summarize how often and how stably an individual engages particular network configurations. These features have been linked to aging, cognition, and neuropsychiatric disorders, suggesting that they capture biologically meaningful aspects of brain function that extend beyond static connectivity.

To date, however, it is unknown whether dynamic rs-fMRI brain states can effectively predict quantitative amyloid burden. In this work, using ADNI data, we test whether subject-level HMM estimates—especially FO and related temporal metrics—derived from rs-fMRI can predict individual amyloid PET SUVR. By integrating dynamic brain state information with molecular imaging, we aim to evaluate HMM-based rs-fMRI measures as non-invasive surrogate markers of amyloid pathology.

Methods

Data

Resting fMRI scans are selected from ADNI phase 2 and phase 3 [2]. The selection criteria are TR = 3s, resting fMRI with a time length of 192, and at least one PET scan performed within 6 months. In total, 621 scans were selected with the corresponding nearest PET scan results. All subjects underwent preprocessing, including slice-time correction, realignment, and normalization to MNI space. The fMRI data were denoised by regressing out CSF and gray-matter signals and then parcellated into 32 regions using the CONN-32 atlas. For each region, the meantime series was extracted, yielding a 32-dimensional representation. The amyloid value, referred to as the summary SUVR or AV45, was selected.

Hidden Markov modeling of rs-fMRI

To characterize dynamic brain states, we fit a group-level Hidden Markov Model (HMM) with Gaussian emissions using the HMM-MAR toolbox in MATLAB. Subject-specific $T \times p$ matrices were concatenated along the temporal dimension, and a vector of segment lengths was used to preserve subject boundaries. We specified an HMM with 12 hidden states, order 0 (Gaussian, non-autoregressive observation model), full covariance matrices, and non-zero state means. Model initialization and estimation followed the default variational Bayes procedure in HMM-MAR with multiple random restarts; the run with the highest free energy was retained. For each time point, the algorithm yielded posterior state probabilities (Gamma) and a most likely state sequence (Viterbi path).

Subject-level HMM features

For each subject, we derived three classes of HMM features:

- 1.Fractional occupancy (FO):** the proportion of time points assigned to each of the 12 states, computed from posterior probabilities (12 features).
- 2.Mean dwell time:** the average length (in seconds) of contiguous visits to each state, estimated from the Viterbi path (12 features).
- 3.Transition probabilities:** a subject-specific 12×12 transition matrix, obtained by counting transitions between successive Viterbi states and normalizing rows to sum to one (144 features).

All features were concatenated into a subject-level predictor vector, yielding a design matrix X of size $N \times p_{feat}$, where N is the number of subjects and p_{feat} the number of features.

Random forest prediction of amyloid SUVR

We used Random Forest (RF) regression to predict continuous global amyloid SUVR from the HMM-derived features. RF models were implemented in MATLAB using TreeBagger with regression mode, a fixed number of trees, and default settings for other hyperparameters. Given the pilot nature of this study and limited sample size, we did not reserve a separate test set; instead, we focused on in-sample fit and internal performance indices. Model performance was summarized using the coefficient of determination (R^2), adjusted R^2 , and out-of-bag (OOB) error estimates provided by the RF. Predictor importance was quantified using OOB permuted predictor importance to identify which HMM features (FO, dwell time, or specific transitions) contributed most strongly to explaining variance in amyloid SUVR. We explored the following options as possible sets of predictors - all features (168), FO & dwell time (24), FO only (12) and dwell time only.

Results

We started by using all 168 features as predictors. The out of box mean squared error was 0.06. A high raw R-squared value of 0.68 was achieved. But due to the large number of features, the adjusted R-squared value was reduced to 0.57. We have also presented the Variable Importance plot of these 168 features (Figure 1), where the first twelve are the FO, the next twelve are the dwell times and the rest are the transition probabilities. It is evident that the FO of the twelve states play the dominant role.

We next ran the model excluding the transition probabilities (due to its high dimensionality) and keeping the other twenty-four features. The out of box mean squared error was again 0.06. A slightly lower raw R-squared value of 0.64 was achieved. The adjusted R-squared value was increased to 0.63.

Finally, we ran the model using FO only and obtained a raw R-squared value of 0.63 was achieved. The adjusted R-squared value was increased to 0.62.

Since we are not performing any cross validation for the pilot project, it is preferable to minimize the risk of overfitting. Considering the fact that FO only performs practically the same as FO & dwell time, probably FO only model is the best model.

Conclusions

Using subject-level HMM summaries as predictors in a random forest model, we found that the time an individual spends in specific recurrent brain states—as captured by FO—is strongly related to their amyloid burden, and that additional temporal complexity (dwell time, transitions) provides limited incremental value in this dataset.

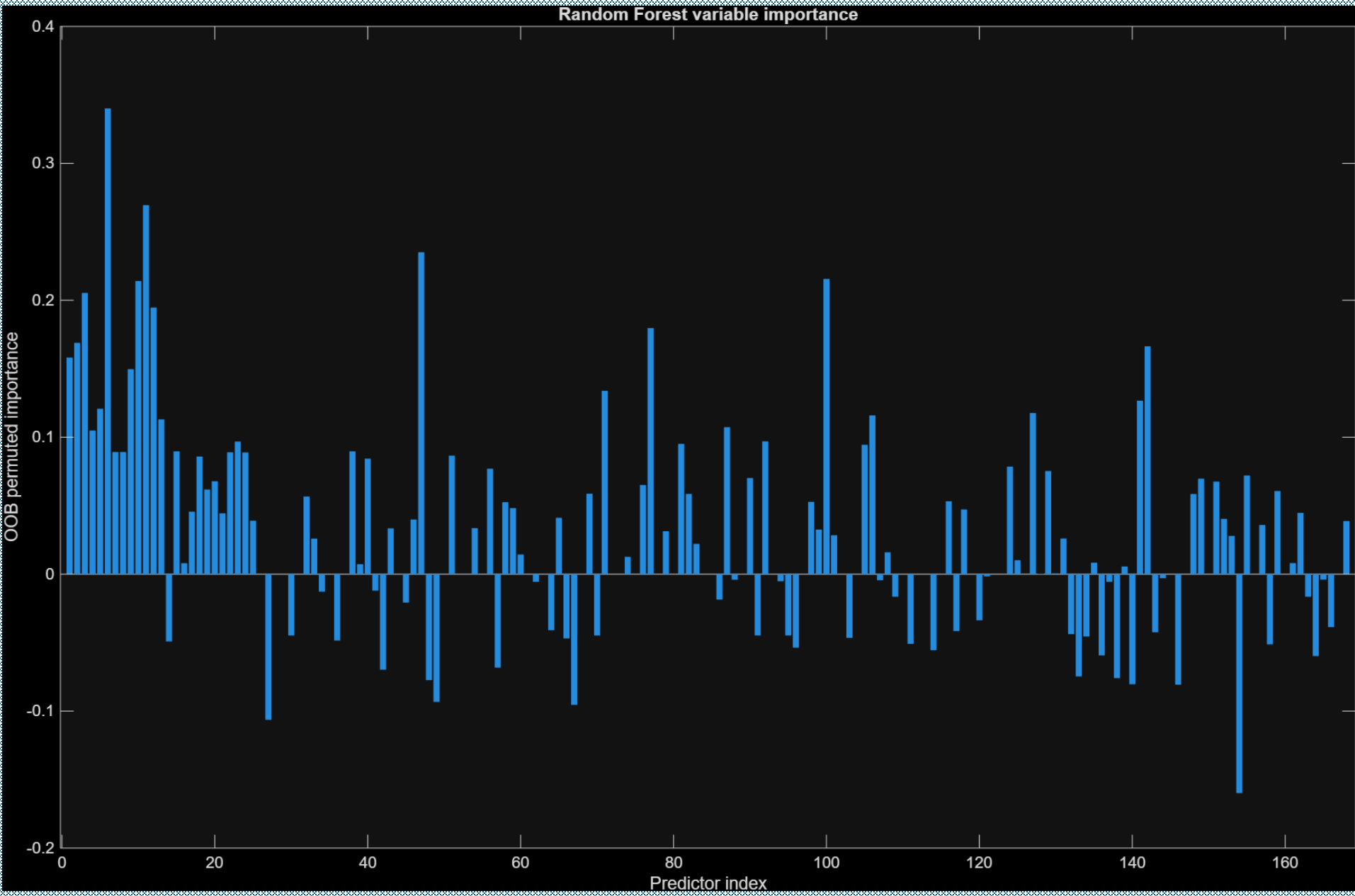


Figure 1. Variable importance plot of the 168 features considered as predictors. the first twelve are the FO, the next twelve are the dwell times and the rest are the transition probabilities.

The pattern that FO-only models perform almost as well as, or slightly better than, models including dwell time and transition probabilities points to FO as a particularly robust and informative summary of dynamic rs-fMRI. This is appealing both conceptually and practically: FO offers a low-dimensional, interpretable representation of an individual’s dynamic repertoire and reduces the risk of overfitting relative to high-dimensional transition matrices. Together, these findings provide preliminary support for the idea that state occupancy profiles derived from HMMs could serve as non-invasive markers of amyloid-related network disruption.

At the same time, these results must be interpreted with caution. This analysis is explicitly exploratory and based on a modest sample size, with no held-out test set or cross-validation. As a consequence, the reported adjusted R-squared values should be viewed as upper bounds on true predictive performance. Potential confounders and site effects were not formally modeled in the prediction framework. Taken together, these factors mean that our findings are hypothesis-generating rather than definitive.

Future work will focus on strengthening and extending these observations. Methodologically, the next step is to implement rigorous evaluation using nested cross-validation and, when possible, external replication datasets, to obtain unbiased estimates of generalization performance. On the HMM side, we plan to examine the impact of varying the number of states, testing alternative observation models (e.g., connectivity-based HMMs), and evaluating test–retest reliability of FO and related metrics.

From a biological and clinical perspective, future analyses will aim to localize which specific states and networks drive prediction, by linking high-importance FO features back to their spatial patterns and putative cognitive relevance. Extensions could incorporate additional modalities such as structural MRI, tau PET, or cognitive measures to build multimodal models of amyloid burden, and to test whether dynamic rs-fMRI features improve classification of amyloid positivity or risk stratification for clinical trials. Ultimately, larger, well-validated studies will be needed to determine whether HMM-derived rs-fMRI metrics can serve as practical, non-invasive surrogates for amyloid PET SUVR in research and, eventually, in clinical workflows.

References

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